

## Algebra II - End-Term - 100 Marks

April 25, 2025: Time: Three hours

1. Let  $p > 0$ . A group  $G$  is a  $p$ -group if the order of every element is  $p^n$ . Show that if  $|G|$  is a power of  $p$ . (5 marks)
2. Let  $G$  be a finite group and  $|G| = p^e m$  where  $e \geq 1$  and  $p \nmid m$ . Let  $H$  be a subgroup of  $G$  and  $|H| = p^a$ ,  $1 \leq a \leq e$ . Show that

$$[N_G(H) : H] \equiv [G : H] \pmod{p}.$$

(5 marks)

3. Let  $G$  be a group and let  $H$  be a subgroup such that  $|G/H| < \infty$ .
- (a) Show that there exists a subgroup  $N \subseteq H$  such that  $|G/N| < \infty$  and  $gNg^{-1} = N$  for all  $g \in G$ . (5 marks)
  - (b) Give an upper bound for  $|G/N|$ . (5 marks)
  - (c) let  $G = D_\infty := \langle s, r \mid s^2 = e, srs = r^{-1} \rangle$ .
    - i. Can  $G$  be written a semi-direct product of two proper subgroups? (8 marks)
    - ii. Determine the normal subgroup  $N \subseteq H$  such that  $|G/N| < \infty$ . Determine  $|G/N|$ . (5 marks)
    - iii. Is  $G/N$  abelian? (5 marks)
4. Let  $G$  be a finite group and let  $G = n$  and let  $p$  be a prime which divides  $n$ .
- (a) Suppose every Sylow- $p$  subgroup of  $G$  is normal. Show that  $G$  is a direct product of its Sylow subgroups. (7 marks)
  - (b) Suppose  $G$  is abelian.
    - i. Describe all the Sylow  $p$ -subgroups of  $G$ . (7 marks)

- ii. Prove or disprove that  $G$  is a direct product of its Sylow  $p$ -subgroups. (5 marks)
5. Let  $A$  be an  $n \times n$  real matrix.
- State the definition of a positive definite matrix. (3 marks)
  - Prove that if  $A$  can be expressed as  $A = TT^T$ , where  $T$  is a lower triangular matrix with strictly positive diagonal entries, then  $A$  must be positive definite. (7 marks)
6. Let  $V$  be an inner product space and  $\{v_1, v_2, \dots, v_n\} \subset V$ . The Gram matrix  $G$  is defined by  $G_{ij} = \langle v_i, v_j \rangle$ .
- Prove that  $G$  is a symmetric positive semi-definite matrix. (7 marks)
  - Prove that the set  $\{v_1, \dots, v_n\}$  is linearly independent if and only if  $\det(G) > 0$ . (7 marks)
7. Let  $G$  be a group of order 12 and  $K$  a Sylow-3 subgroup. Consider  $G/K = \{g_1K = K, g_2K, g_3K, g_4K\}$  and let  $Sym(G/K)$  denote the symmetric group on the set  $G/K$ . Let  $\phi : G \rightarrow Sym(G/K)$  be the map given by  $\phi(g)(g_iK) := gg_iK$ .
- Show that  $\phi$  is a homomorphism. (5 marks)
  - Show that  $\phi$  is injective if and only if  $K$  is not a normal subgroup of  $G$ . (7 marks)
  - If  $\phi$  is injective, show that  $G \cong A_4$ . (7 marks)